

Adopt source_location from Library Fundamentals V3 for C++20

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1 Class source_location [reflection.src_loc]

1.1 Header <source_location> Synopsis [reflection.src_loc.intro]

```
namespace std {  
    struct source_location {  
        constexpr source_location() noexcept;  
  
        constexpr uint_least32_t line() const noexcept;  
        constexpr uint_least32_t column() const noexcept;  
        constexpr const char* file_name() const noexcept;  
        constexpr const char* function_name() const noexcept;  
  
        static consteval source_location current() noexcept;  
    };  
}
```

[*Note:* The intent of `source_location` is to have a small size and efficient copying.– *end note*]

```
constexpr source_location() noexcept;
```

1 *Effects:* Constructs an object of class `source_location`.

2 *Remark:* The values are implementation-defined.

```
constexpr uint_least32_t line() const noexcept;
```

3 *Returns:* The presumed line number (16.8) represented by this object.

Remark: Line numbers are presumed to be 1-indexed however an implementation is encouraged to return 0 when the line number is unknown.

```
constexpr uint_least32_t column() const noexcept;
```

4 *Returns:* An implementation-defined value representing some offset from the start of the line represented by this object.

Remark: Column numbers are presumed to be 1-indexed however an implementation is encouraged to return 0 when the column number is unknown.

```
constexpr const char* file_name() const noexcept;
```

5 *Returns:* The presumed name of the current source file (14.2) represented by this object as an NTBS.

```
constexpr const char* function_name() const noexcept;
```

6 *Returns:* If this object represents a position in the body of a function, returns an implementation-defined NTBS that should correspond to the function name. Otherwise, returns an empty string.

```
static constexpr source_location current() noexcept;
```

7 *Returns:* When invoked by a function call whose *postfix-expression* is a (possibly parenthesized) *id-expression* naming **current**, returns a **source_location** with an implementation-defined value. The value should be affected by **#line** (14.4) in the same manner as for **__LINE__** and **__FILE__**. If invoked in some other way, the value returned is unspecified.

8 *Remark:* When a *brace-or-equal-initializer* is used to initialize a non-static data member, any calls to **current** should correspond to the location of the constructor or aggregate initialization that initializes the member.

9 [*Note:* When used as a default argument (9.3.6), the value of the **source_location** will be the location of the call to **current** at the call site. – *end note*]

[*Example:*

```
struct s {
    source_location member = source_location::current();
    int other_member;
    s(source_location loc = source_location::current())
        : member(loc) // values of member will be from call-site
    {}
    s(int blather) : // values of member should be hereabouts
        other_member(blather)
    {}
    s(double) // values of member should be hereabouts
    {}
};

void f(source_location a = source_location::current()) {
    source_location b = source_location::current(); // values in b represent
this line
```

```
}

void g() {
    f(); // f's first argument corresponds to this line of code

    source_location c = source_location::current();
    f(c); // f's first argument gets the same values as c, above
}

– end example ]
```

2 Feature macro

We recommend the feature macro `__cpp_lib_source_location` for this feature.